

Summer Internship Summary

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08/13/2026

Overview

A somewhat chronological ordering:

1) First project

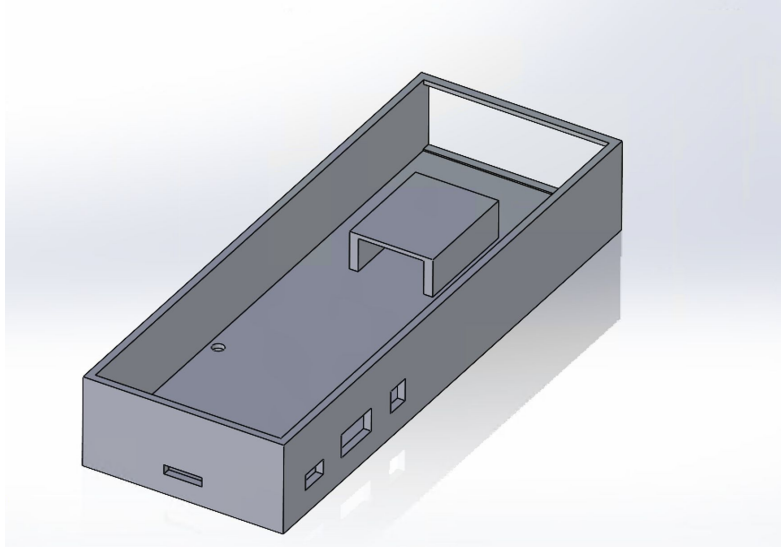
2)  QKD

3) 

4) Miscellaneous

Raspberry Pi + RF Module Case

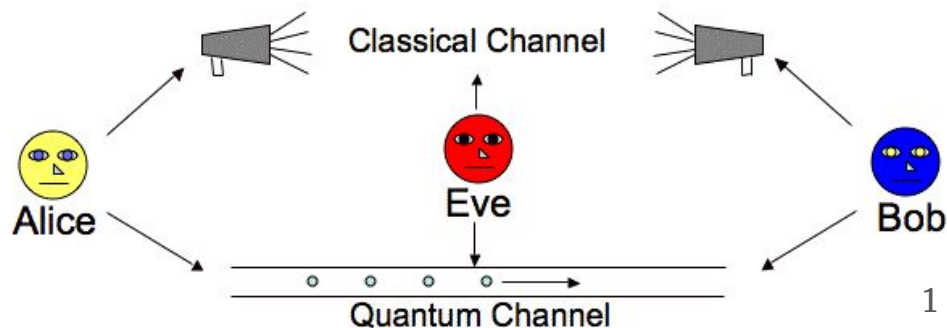
Learning CAD on Solidworks:



Introduction to

Main goals:

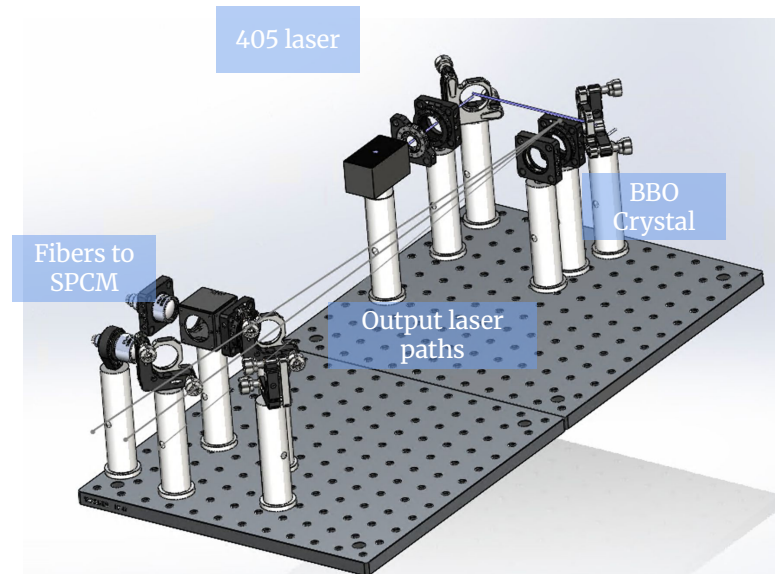
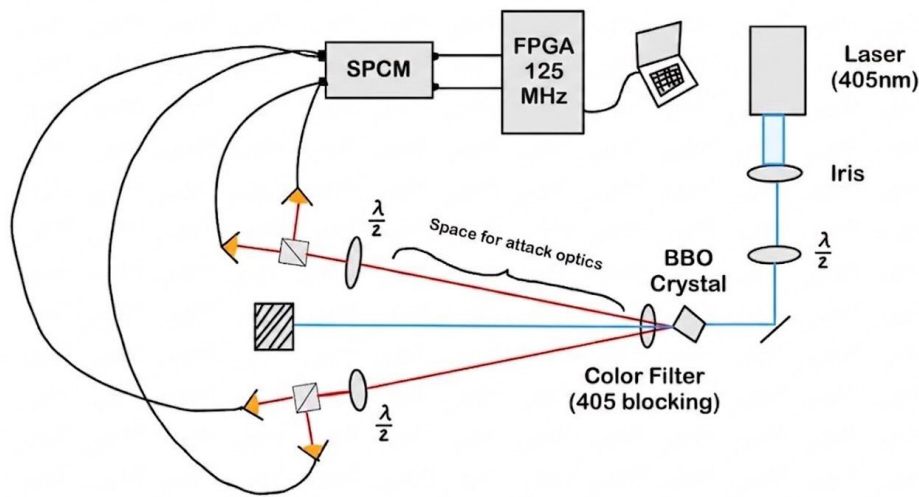
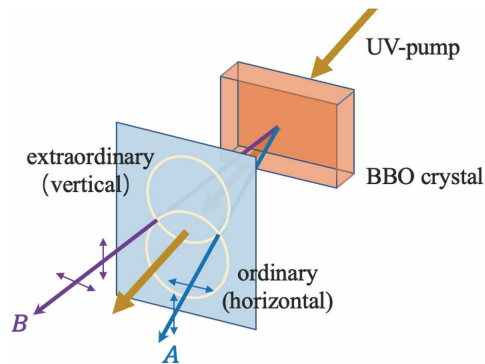
- BB84 Quantum key distribution is theoretically infallible ... does this hold up in real life?
 - Create testing software for verifying the outcomes of a transmission
 - **Create hardware to verify the new software**
 - Normal transmission
 - Attacks from a 3rd party
 - Mechanical failures



Experiment Design

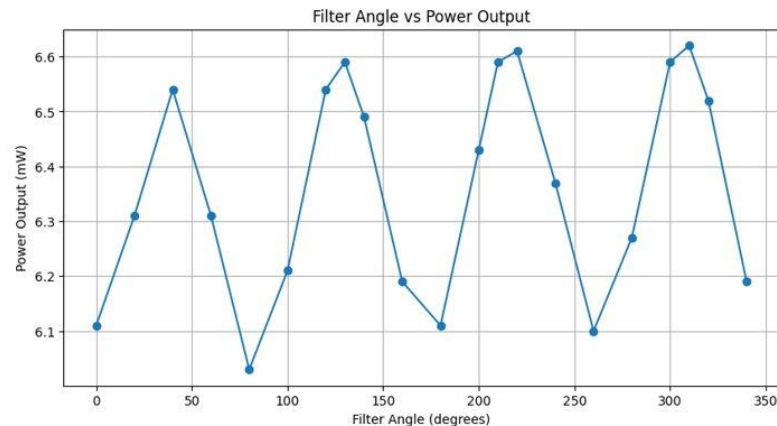
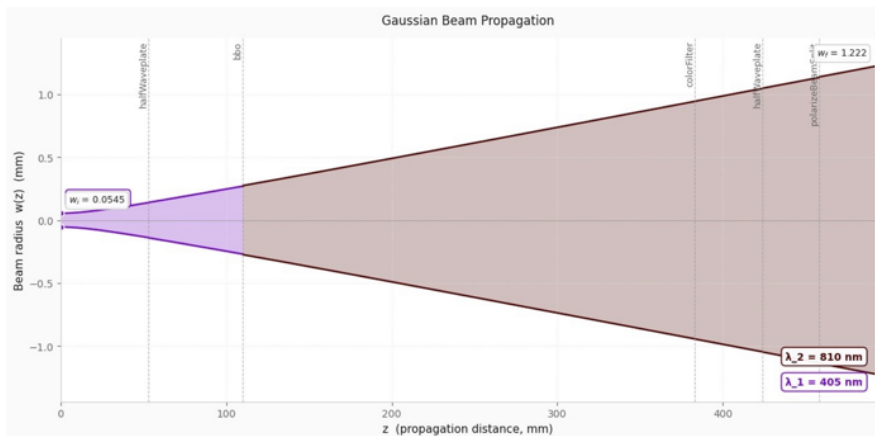
Read various papers [2][3][4] to arrive at this experiment design:

- BBO (entanglement generating crystal, 405nm $\rightarrow$ two 805nm)
- FPGA for single photon counting from fibers



Building the Experiment

- Took inventory of current parts and ordered needed ones
- Made calculations for laser
 - Propagation distance needed
 - Power and beam spot size
- Built initial experiment to ensure BBO response

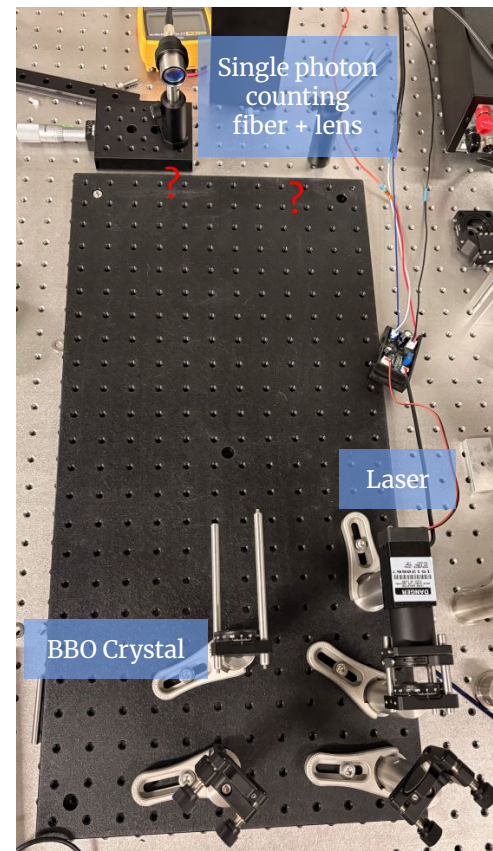


Building the Experiment

- Currently able to send light through BBO
- Where are the expected 810nm entangled beams?
 - Power meter ✗
 - Avalanche photodiode ✗

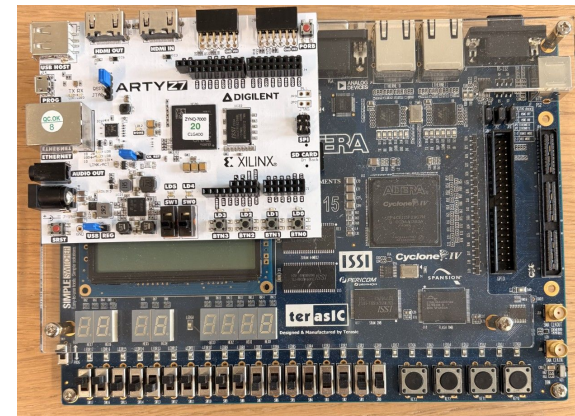
Calculated W/mm^2 : 60–600 fW

⇒ Need **single photon detection (SPD)** via FPGA



Programming the FPGA

- First try: ALTERA DE2-11
 - Easier for learning, bad for getting access to data via software
- Final solution: ARTY Z7 with a ZYNQ processor
 - Allows for easy software access via PYNQ images



Simple FPGA program:

```
import hardware constraints
```

```
monitor trigger on GPIO_in
```

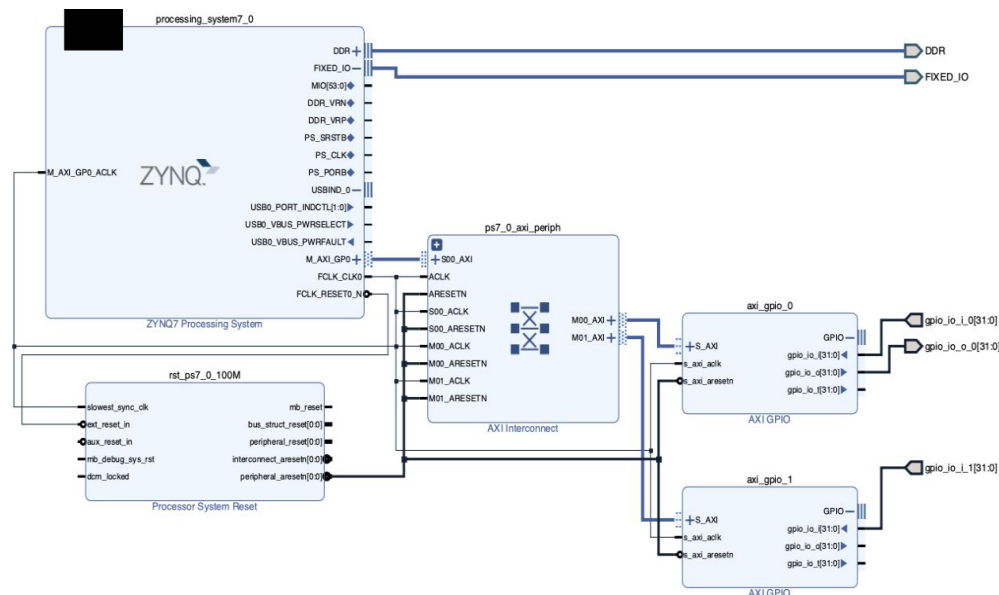
```
if trigger on:
```

```
    count rising edges from GPIO_in
```

```
    write count to GPIO_out
```

```
if trigger off:
```

```
    stop counting
```



Software Control of FPGA

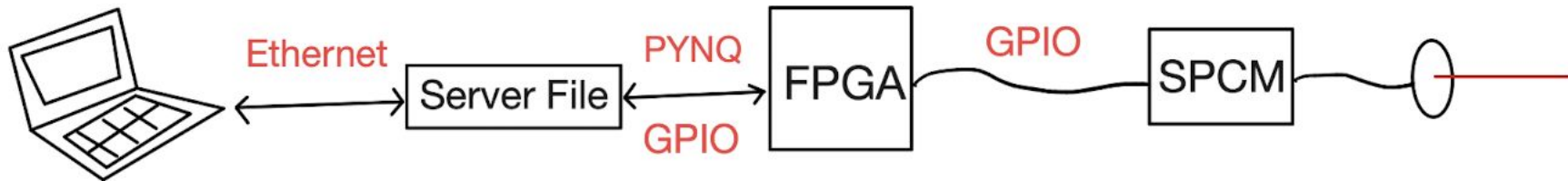
Front end: Python GUI

- Timer for desired measurement time, or manual
- Triggers and reads counting from GPIO via websocket

Back end: Python web server

- Create web server and socket port for front end
- Access the FPGA GPIO via PYNQ

GUI File



Next Steps

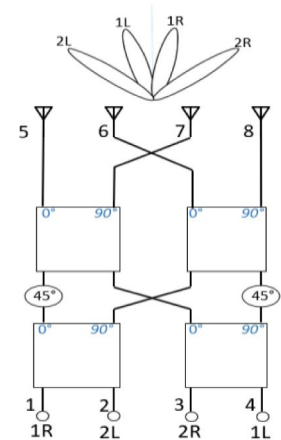
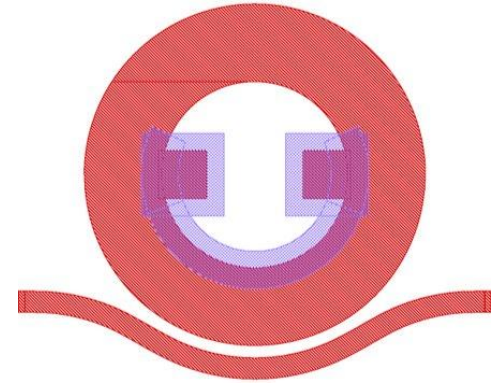
- Verify that the red entangled beam is where we expect
- Finish setting up experiment for full testing
 - Expand FPGA control for four SPCM channels
 - Test correlations between entangled photons
- Coordinate with Menlo Park Team to test attack scenarios



Overview

Main goals:

- Optical phased arrays using butler matrices, on a PIC (photonic integrated circuit)
 - 2D Optical beam steering (transmitting)
 - Beam receiving at array of angles
- Small form factor
- Automation of setup and controls



My work on



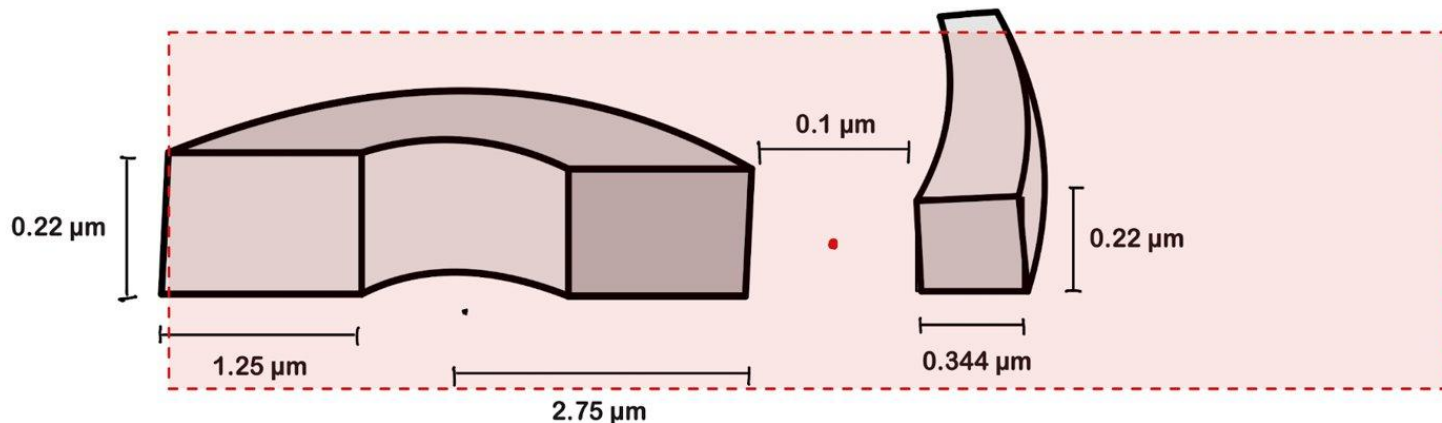
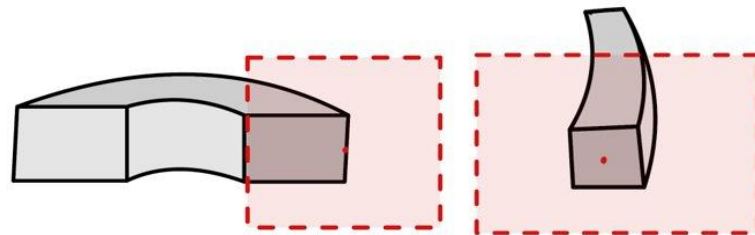
- Simulations and numerics
 - Phase shifting ring resonator simulations
 - Q-factor analysis
- Software controlled PIC testing for optical beam steering



Phase Shifter Simulations

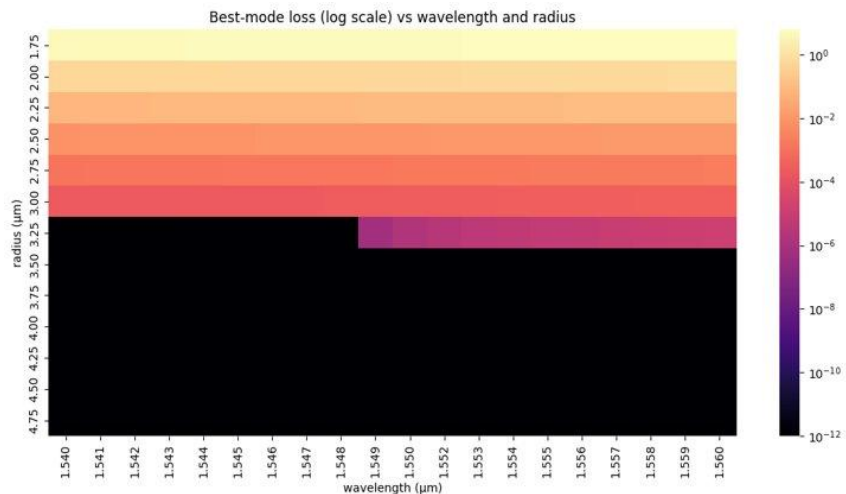
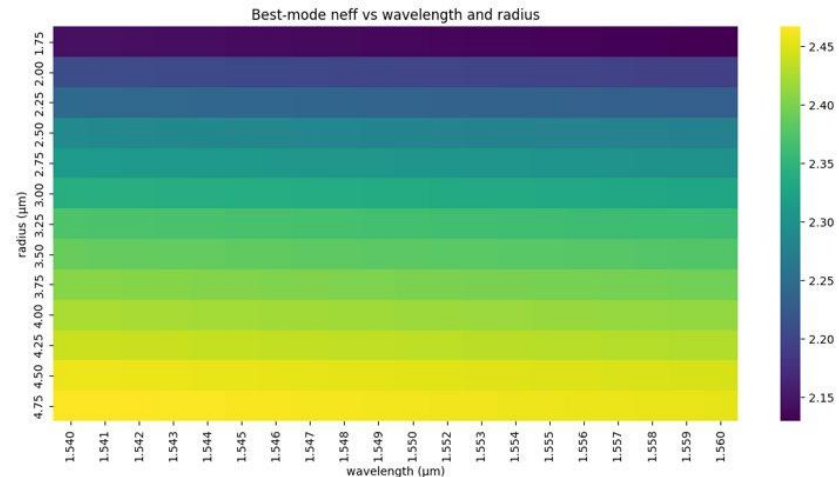


- Learned to use Tidy3D
- Three different simulations:
 - Sweep over **wavelengths**:
 - Ring resonator only, **sweep over radii**
 - Bus waveguide only, **sweep over widths**
 - Full ring and bus, checking coupling for resonant wavelengths



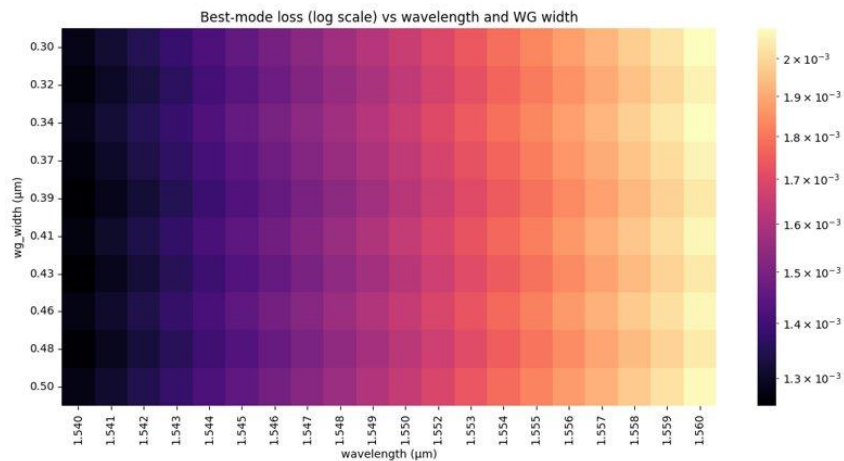
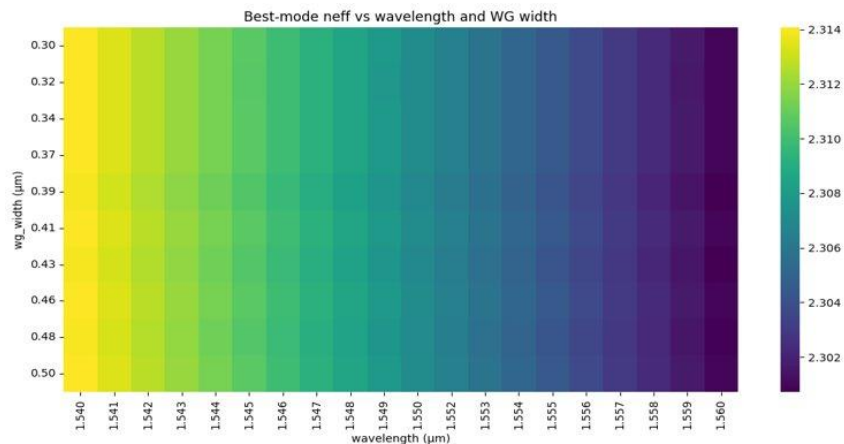
Ring Resonator Simulations

- Swept wavelength between 1.54 and 1.56 μm , in anticipation of setpoint of 1.55 μm
- Swept radii between 1.75 and 5 μm
- Solved for the top 15 modes, picking out the best mode at each step



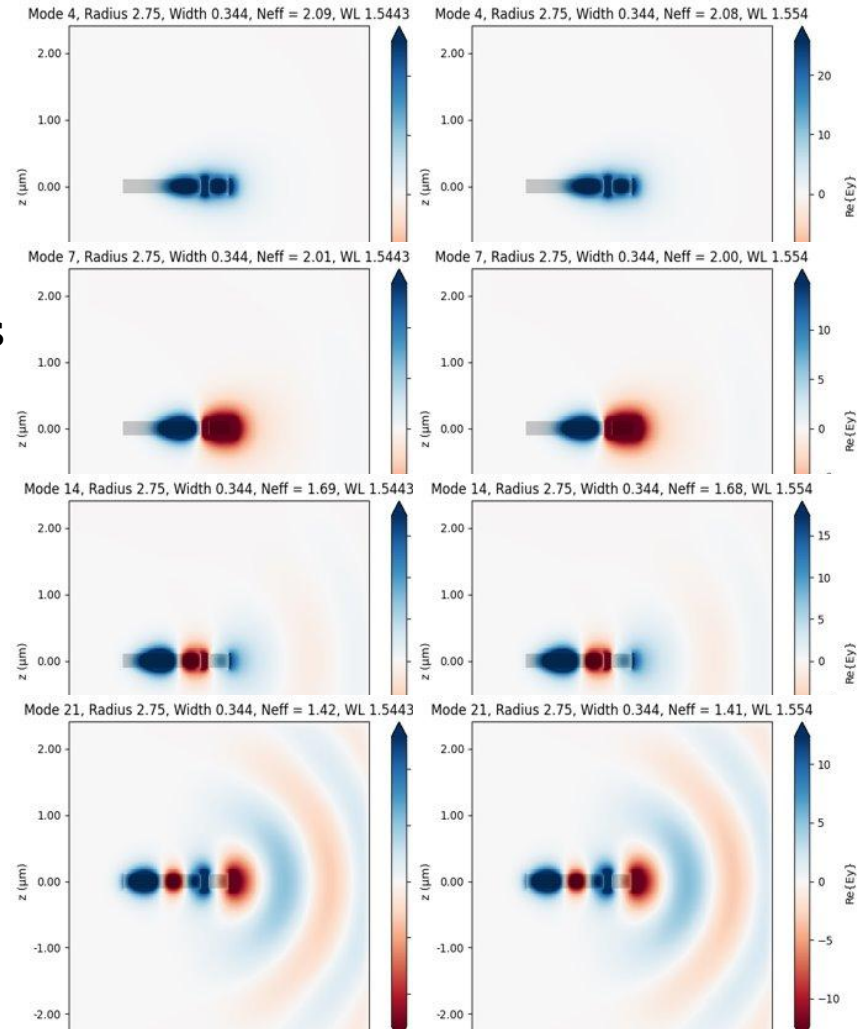
Bent Waveguide Simulations

- Swept wavelength between 1.54 and 1.56 μm , in anticipation of setpoint of 1.55 μm
- Swept width between 0.3 and 0.5 μm
- Solved for the top 15 modes, picking out the best mode at each step



Full System Simulations

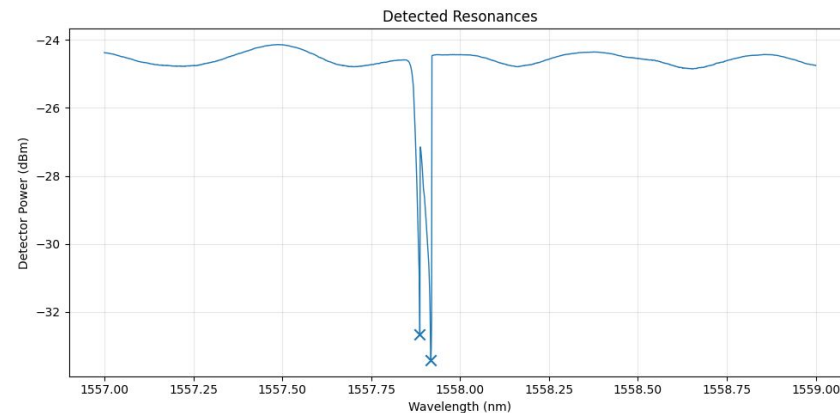
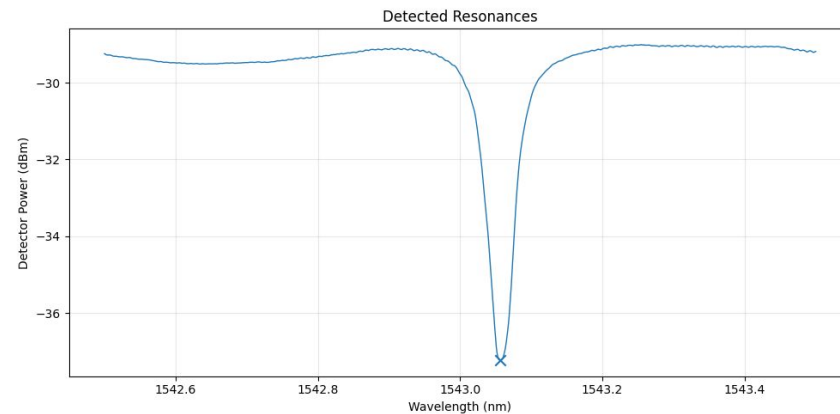
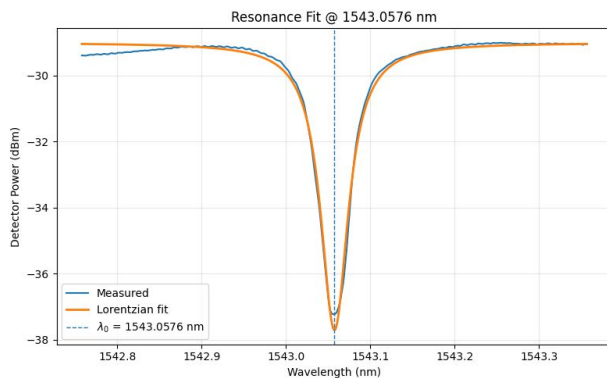
- Checked coupling at **1.554 μm** (desired) and **1.5443 μm** (undesired) resonant wavelengths
- Found **four** physical coupling modes, with one pair of **anti-symmetric** modes
- Modes and effective index (n) almost exactly the same between both wavelengths



Q-Factor Analysis

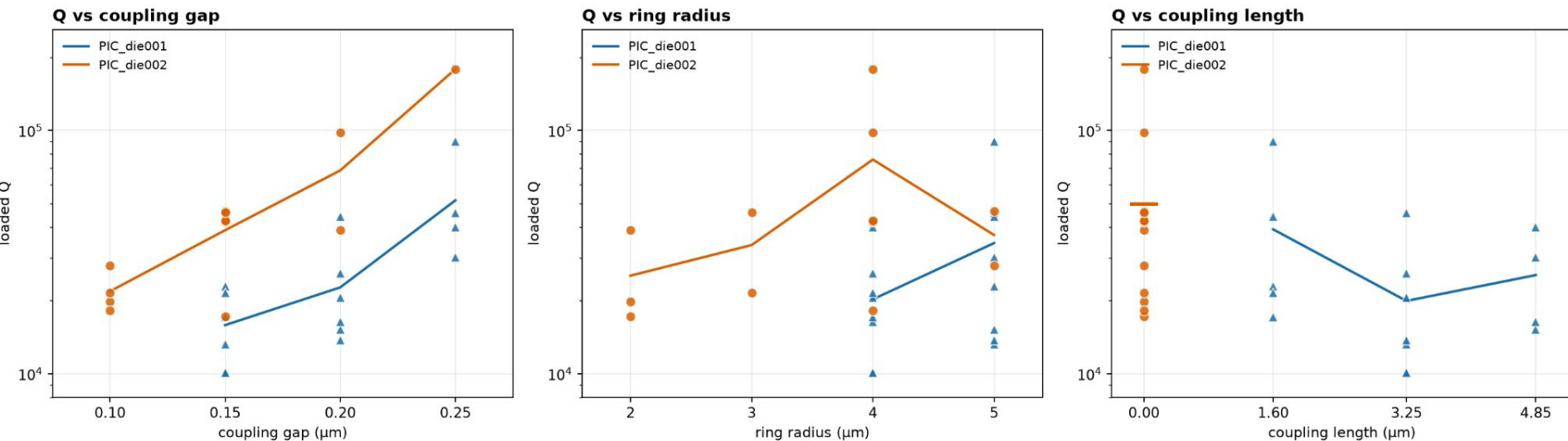


- Found resonant peaks vs wavelength for sweeping:
 - Radius
 - Ring resonator/bus gap
 - Coupling length
- Filtering out “bad” peaks



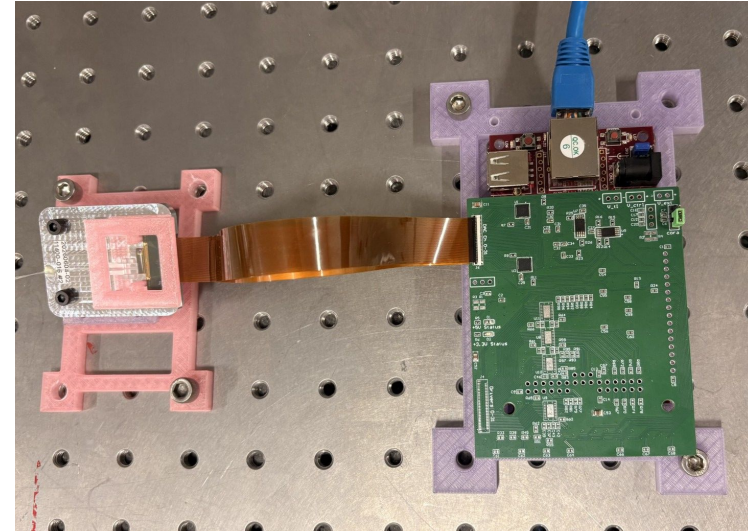
Q-Factor Analysis

- Relationship only existed for Q-Factor vs coupling gap

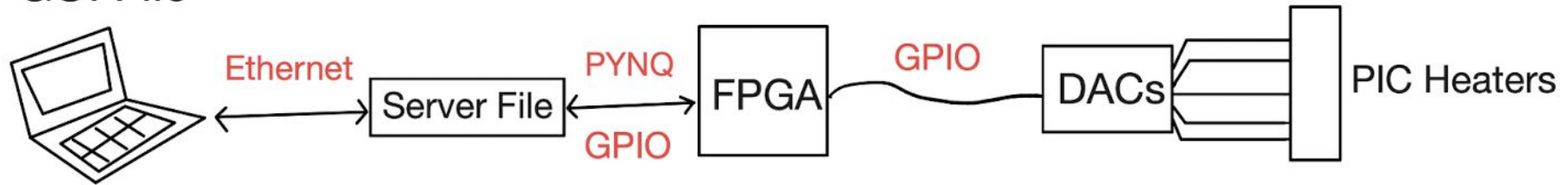


Testing PICs with Software Control

- Created a GUI for controlling and testing optical beam steering on PICs
 - Easier testing
 - Choice of channel or angle control
 - 64 channel ability



GUI File



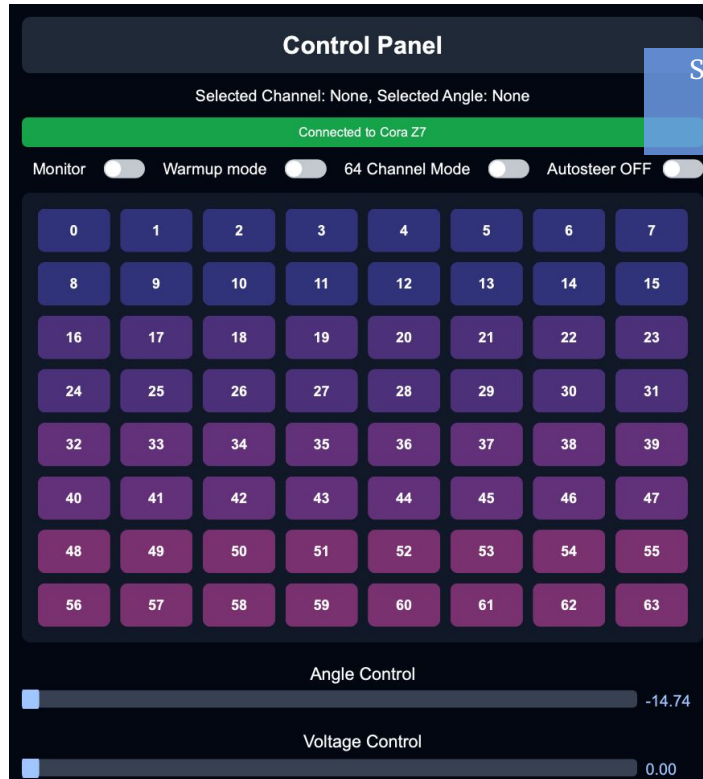
Python GUI

Warmup
mode

PIC Channel
Select

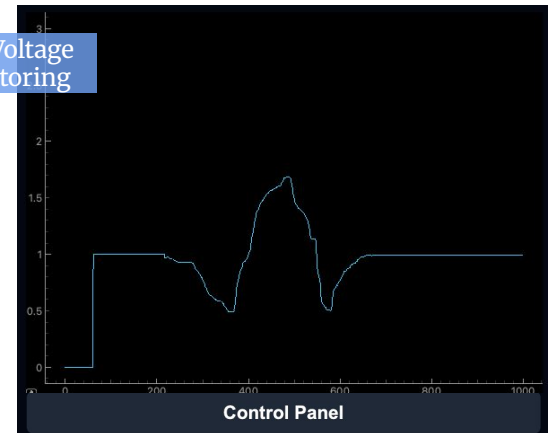
Beam Angle
Select

DAC Voltage
Control



Software cued
auto beam
steering

Live Voltage
Monitoring

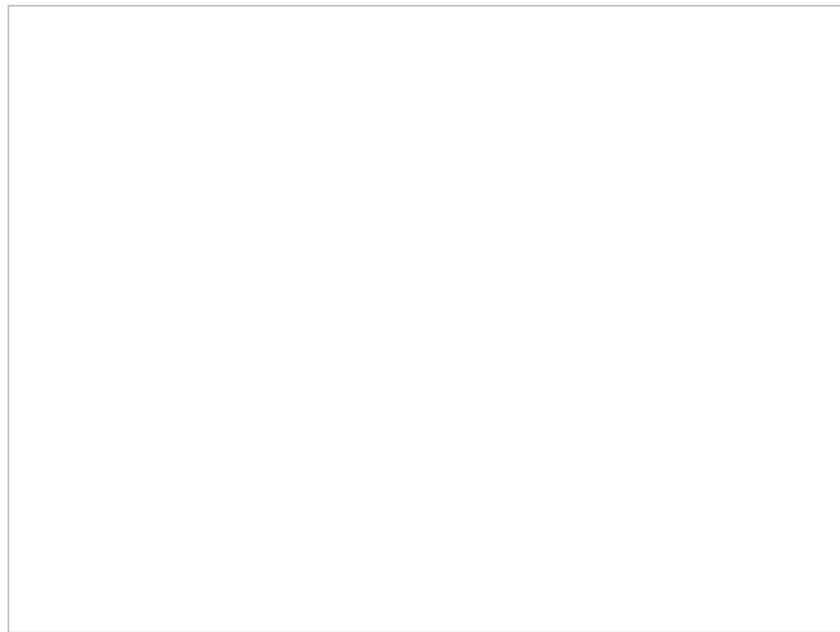


Dual 32
channel mode



Testing and Steering

- Testing:
 - Set voltage for each channel/angle
 - Voltages automatically recorded and saved by GUI
 - Saved voltages, resistances and wavelengths in sharepoint
- Auto steering:
 - **Software controlled:** limited by Python and ethernet transfer speeds
 - **Software cued:** untested, FPGA controlled for very fast auto steering



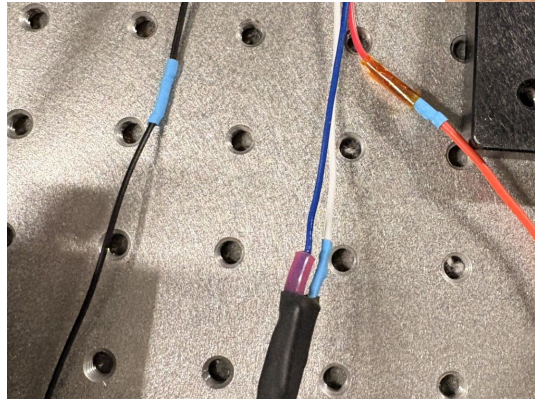
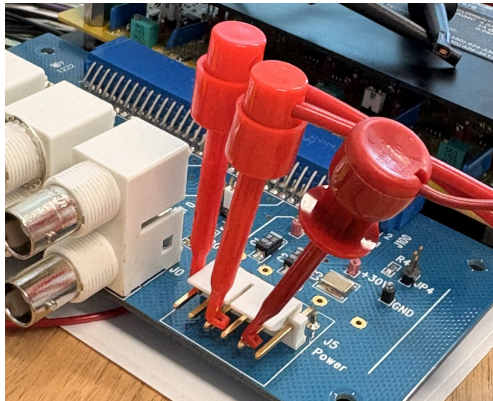
Next Steps

- Verify fast (10 μ s per channel) auto steering via software-cued FPGA
- Test dual butler matrix mode for 2-axis control
- Auto-calibration via photodiodes
- Phase masking row-column address control via GUI



Along the way

- Soldering wires and connectors
- Prepping PCBs
- Debugging and troubleshooting
- Lots of learning!



Thank you everyone!

Questions?

Sources

Feng, Tianxuan, et al. "Low-Power System Model for Quantum Entangled Photon-Pair Source." *Journal of Systems Engineering and Electronics*, vol. 35, no. 5, 2024, pp. 1287–94. ResearchGate, <https://doi.org/10.23919/ISEE.2024.000104>.

Haitjema, Mart. "A Survey of the Prominent Quantum Key Distribution Protocols." Washington University in St. Louis, 2 Dec. 2007, www.cse.wustl.edu/~jain/cse571-07/ftp/quantum/.

Phu-arphit, Jeerasak, et al. "An Efficient Method for Generating Quantum Entangled Photon Pairs." *Procedia of Multidisciplinary Research*, vol. 3, no. 7, 2025, p. 46, so09.tci-thaijo.org/index.php/PMR/article/view/6982.

Yang, Yiqian, and Liangcai Cao. "Quantum Hyper-Entangled System with Multiple Qubits Based on Spontaneous Parametric Down-Conversion and Birefringence Effect." *Optical and Quantum Electronics*, vol. 56, no. 1, 2024, article 12, <https://doi.org/10.1007/s11082-023-05590-2>.

Zhou, Xiaoyan, et al. "Compact Quantum Entangled-Photon Source for Space Platform." *Acta Optica Sinica*, vol. 42, no. 3, 2022, article 0327008, <https://doi.org/10.3788/AOS202242.0327008>.